Bay Area Real Estate Prediction Model & Dashboard

Team 211 · CSE 6242
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INTRODUCTION / MOTIVATION

The Problem/Introduction

Buying a home in the Bay Area is one of the most complex financial decisions a family can make.

Median prices $1.15M – $1.50M across Santa Clara, San Mateo, and Alameda counties

Price variation >40% across zip codes within the same county

Buyers must juggle property attributes, neighborhood quality, affordability constraints, and market timing which are currently scattered across Zillow, Redfin, and city databases with **no unified decision-support tool.** We try to solve this unmet need with our interactive dashboard product.

MACHINE LEARNING APPROACH

5 regression models trained on 80/20 split across around 50,000 Bay Area property records. Our prediction function accepts a street address, geocodes it, engineers the relevant features identically to training, and returns a **Gradient Boosting**-predicted price.

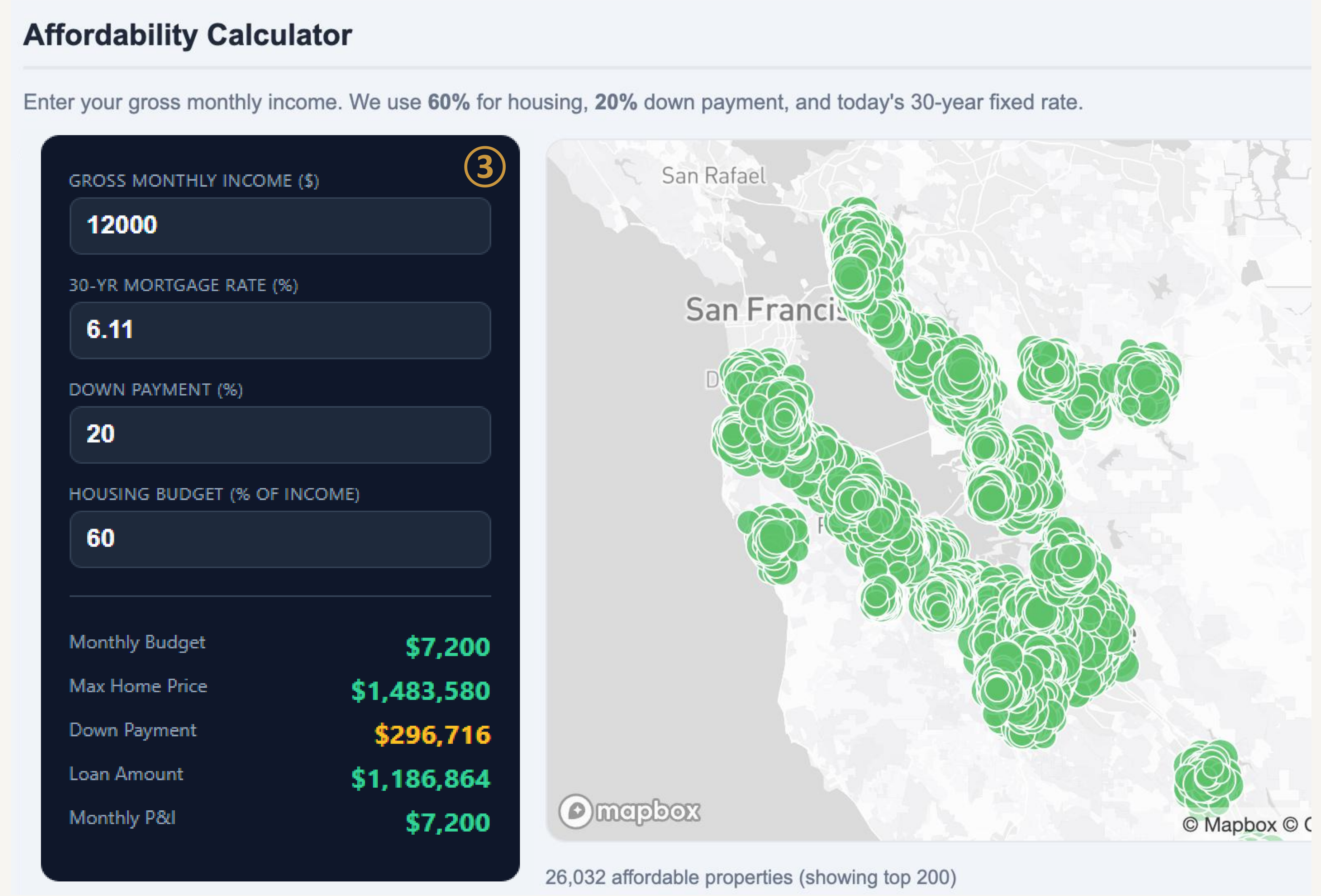
Model	R^2	MAPE	RMSE
Gradient Boosting ★	0.927	11.3%	$239K
XG Boost	0.927	11.4%	$241K
Random Forest	0.922	11.6%	$248K
Ridge Regression	0.899	15.9%	$283K
Linear Regression	0.899	15.9%	$283K

Table: Different ML model performance

INTERACTIVE D3.JS DASHBOARD

Dual-layered visualization transitioning from regional affordability trends to individual property analysis:

1. **Coolness & Kid Friendly** filterability for neighborhood needs
2. **Interactive tooltips** with Kid-Friendly & Cool score bar charts on hover
3. **Affordability Calculator** providing cost realistic filtering with income & downpayment chosen
4. **Property-level predictions** via Gradient Boosting backend for any Bay Area address



SUMMARY & CONCLUSIONS

This project proves that housing search can be improved through an integrated, decision-oriented framework that combines price prediction, affordability analysis, and neighborhood quality assessment within a single interactive dashboard. The dashboard translates these modeling results into a practical visual tool, allowing users to explore Bay Area properties by budget, Coolness, and Kid-Friendly score across 143 zip codes while comparing homes in a more personalized and interpretable manner.

Using approximately 50,000 property records, the model achieved strong predictive performance, and the proposed neighborhood scores contributed additional explanatory value beyond traditional listing platforms. Together, these components illustrate how data-driven, user-centered design can support more informed housing decisions in competitive real estate markets.

DATA

5 sources · 3 Bay Area counties · 143 zip codes

Zillow Research
Property-level sale prices

OpenStreetMap (OSMnx)
park polygons

WalkScore API
Walk/Bike/Transit scores

Redfin Market Tracker
Aggregate market stats

TripAdvisor
Restaurants & attractions

Data Sourced by Scraping / Downloading

- Around 50,000 properties · 143 zip codes · Calibrated from real aggregate prices
- 29 features trained
- Novel features (kid friendly score, coolness score) engineered and trained
- Bay area real estate sale price for past five years curated

NOVEL FEATURE: KID-FRIENDLY SCORE

No platform today lets buyers filter neighborhoods by a quantifiable family-suitability metric. We queried park polygons from OpenStreetMap via OSMnx and derive a kid-friendly score

Raw park count (MinMax-scaled) — 50%

County-relative percentile rank — 30%

Bay Area overall percentile — 20%

AFFORDABILITY CALCULATOR

DTI-based filtering using a configurable 50% monthly affordability threshold. Buyers input gross income, down payment, and mortgage rate to instantly highlight affordable zip codes.

Example: A $200K household with $150K down payment → max purchase price of ~$612K → covers ~18.5% of properties in dataset.

Gross Annual Income	$200,000
Down Payment	$150,000
Interest Rate (%)	6.5%

→ Max affordable price: $612,000 | Matched zip codes: highlighted on map

EXPERIMENTS & RESULTS

0.927 R^2 (Gradient)

11.3% MAPE

~50K Properties

Performance by price range:
- **Best:** $500K–$2M range (majority of Bay Area transactions)
- **Degraded:** Luxury homes >$3M, under 500K (sparse training data, idiosyncratic pricing)

Evaluation
80/20 split on ~50K properties; metrics include MAPE (primary) and R^2, etc

Results:
Gradient Boosting achieves $R^2 = 0.927$ and MAPE = 11.3%, with 46% of predictions within 10% of true price.
Top features for predicting Bay Area real estate sale price: square footage (77%), sale year (6%), lot size (5.5%), and property type — single family (3.75%). Square footage is the dominant predictor by a wide margin, followed by sale year as a temporal market indicator. Lot size and single-family property type, while contributing modestly, underscore the role of property characteristics and land value in driving sale price.

Comparison:
Ensemble models outperform linear regression by ~4.5 MAPE points, capturing nonlinear pricing effects more effectively.

NOVEL FEATURE: COOL SCORE

Captures walkability and neighborhood vibrancy. Walk Score weighted highest as amenity proxy; Transit Score lowest (Bay Area remains car-dependent outside SF/Oakland).

Walk Score	30%
Restaurant density	25%
Attractions (TripAdvisor)	20%
Bike Score	15%
Transit Score	10%

LITERATURE HIGHLIGHTS

Park & Bae (2015)
Ensemble methods outperform hedonic regression → we extend with Gradient Boosting + geo-features

Baldominos et al. (2018)
Gradient Boosting achieves 5–8% MAPE — consistent with our 11.3% on a harder Bay Area dataset

Crompton (2001)
Parks create 10–20% property premiums → justifies our Kid-Friendly Score as price predictor

Boeing (2017)
OSMnx for OpenStreetMap queries → used to extract 2,900+ park polygons

Blumenberg & King (2024)
Bay Area housing costs drive 60+ min commutes → motivates affordability-first design

CONCLUSION

What we built:
The first unified Bay Area real estate tool combining price prediction, affordability filtering, and neighborhood quality scores in a single interactive D3.js dashboard.

Limitations:
Home predictions degrade above $3M or under 500K; city-level TripAdvisor data does not capture zip-level variation; no formal user study conducted. Listing photographs were excluded from the training features, and more complex deep learning architectures were not explored due to project time constraints.

Future work:
School quality scores, commute time integration, time-series forecasting, formal IRB user study.

Benchmark vs. Zillow Zestimate
30–50 recent Redfin listings with known closing prices tested against both our MAPE and Zillow's published error rate — results reported in final evaluation.